Practical Network Defense

Master's degree in Cybersecurity 2025-26

Course introduction

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Practical Info

The lecturer

- Angelo Spognardi
  - Associate professor, Dipartimento di Informatica
  - Tel: 06 4925 5164
  - https://angelospognardi.site.uniroma1.it
  - spognardi@di.uniroma1.it
- Student hours:
  - On request by email
  - After: my office G29, v.le Regina Elena, 295, Edificio G
- My research interests:
  - Computer network security, security in social networks, ML and AI in cybersecurity, privacy, applied cryptography, operating systems, programming

This course

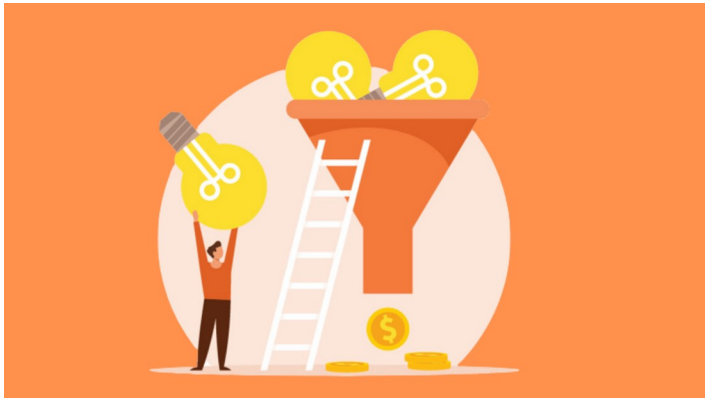
- Website:
  https://sites.google.com/di.uniroma1.it/practical-network-defense
  - Join the class in classroom
    classroom.google.com
    using this code: nctzr6lx
  - Use your @studenti.uniroma1.it account (or it won't work…)
- 6 CFUs
- Schedule:
  - Wednesday 12-14 Aula 3, v.le Castro Laurenziano 7ª
  - Thursday 12-15 Tiburtina Labs (lab XV)

Objectives

- Methods and tools for the protection of computer networks
- Focus on practical application of the concepts learned:
  - protocols in networked computer systems
  - mechanisms commonly used to compromise a computer system's security
  - mechanisms used for the detection of intrusion attempts in computer networks
- At the end of the course students will be able to:
  - monitor traffic in networks
  - apply a security policy
  - perform a network scan and search for vulnerabilities in a computer network
- Students will develop the ability to:
  - select the appropriate firewall rules to protect a network
  - select the most appropriate mechanisms to protect a networked computer system
  - make the most appropriate design choices to implement a "defense in depth" strategy, using isolated networks and dedicated tools (VPN, proxy and firewall)
- Students will learn how to document their choices. They will also have acquired the ability to prepare presentations related to specific scientific topics

Topics covered (tentative)

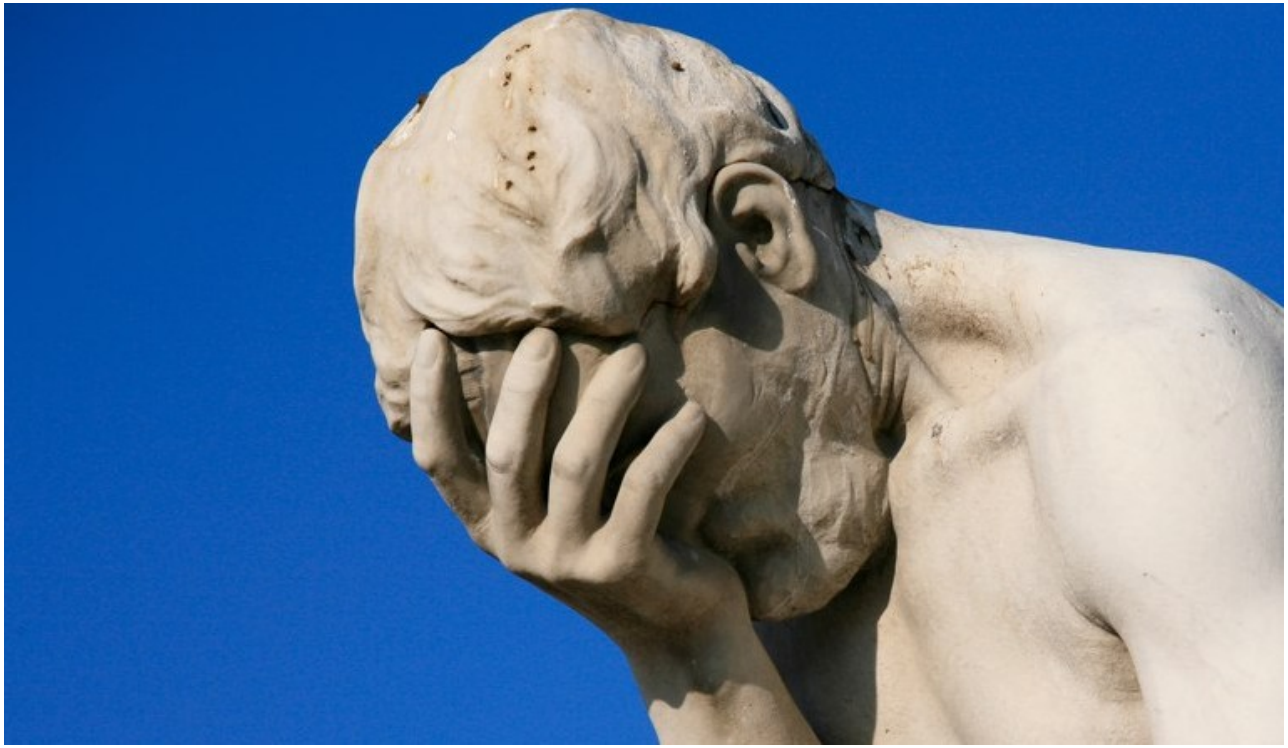
- Network monitoring
- Network traffic analysis
- Network attacks (e.g., session hijacking, man-in-the-middle)
- Minimizing exposure (attack surface and vectors)
- Network hardening
- Network policy development and enforcement
- Defense in depth
- Perimeter networks (DMZs)/Proxy Servers
- Implementing firewalls and virtual private networks (VPNs)
- Implementing IDS/IPS
- Network access control (internal and external)



How the classes are structured

- Theoretical concepts presented mainly during classroom hours, but also during lab sessions
- Practical activity during lab hours
- Some topics will have an assignment with two parts:
  1) An implementing part in the virtual lab environment
  2) A reporting part
- Assignments are part of the exam
  - They are a **pre-requisite** for accessing the written exam

Remember that… things can go bad!!



Especially during labs… Please be tolerant with us ☺️

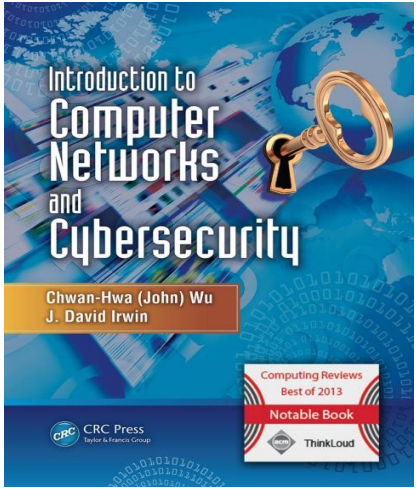
To do the first activities

- We will use Kathará (formerly known as netkit)
  - A container-based framework for experimenting computer networking: http://www.kathara.org/
- A virtual machine is made ready for you: please download it **BEFORE** coming to the labs
  - https://drive.google.com/drive/folders/1ENSXedA5Py0jagl8h_Vr8EuQKm5bHYpe?usp=share_link
- In the following, you will be granted the access to our virtual infrastructure (the ACME co.)

How to pass the exam

- All the rules are applied both to full-time and part-time students
  - 3/4 mandatory assignments
  - full written exam
  - (optional) on demand oral exam for students with a written exam >= 27
- Assignment rules
  - Assignments are individual or group activities to apply techniques and tools introduced during the classes
  - Assignments do not have an hand-in date, but must be handed-in before taking the full-exam
  - Assignments are evaluated A, B or C and provide an increment to the final grade of the written exam (up to +3 points)
- Written exam rules
  - A written exam is valid for one year and supersedes any other written exam done previously

Material



- Hand-notes
- Slides of the lectures
- Articles linked in the website
- Main textbook:
  - **Introduction to Computer Networks and Cybersecurity** (Chwan-Hwa (John) Wu, J. David Irwin, 1e) CRC Press
- Other books will be suggested during the lectures for each considered subject
  - Your contribution is welcome: don't hesitate to share useful resources via classroom

Feedback

- Always welcome
- Write me your suggestions

Hack the box!

- Website with CTF (capture the flag) challenges
- Join our team (and please do your job!)
- To be enrolled:
  - Register yourself in the website
  - Join our telegram group https://t.me/htbsapienza
  - Write in the group your hack-the-box username

Self-assessment!

What is this assessment about?

- If you can not understand some questions at all, you are **definitely lacking** the necessary prerequisites
- If you understand the questions, but just cannot remember the answers, you may think you would look in the book (or would google)…
- You can find the answers in most elementary books on computer security and using your computer (as a engineer would)
- But do not cheat yourself – we will be using the commands and the terms asked about in the test all through the course…
  - Don't get discouraged: simply you have to **fill the gap** to catch up!

Self assessment 1: linux

Self assessment 2: linux2

Self assessment 3: network

Self assessment 4: network2

Self assessment 5: cryptography

That’s all for today

- **Questions?**
- Bonus reference to get used to linux CLI and tools:

  http://overthewire.org/wargames/bandit/bandit0.html

  - Go to bandit and try to reach level 34!!
    - 33 is also good :-)
  - Take notes of the passwords and how you obtained them
  - Try to learn as much as you can solving each level